

BIOTOX TEST WITH MARINE BACTERIA VIBRIO FISCHERI

Group 8

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ABSTRACT

A BIOTOX luminescence inhibition test was performed according to ISO standard 11348-3 to determine the inhibitory effects of 4-nitrophenol on the marine bacterium *Vibrio fischeri*. Bioluminescence was measured using a photomultiplier at t=0, 10, 20 minutes over a concentration range of 7.5 – 240 mg/L, selected around a reference EC50 of 30 mg/L. Duplicate samples were prepared at five concentrations alongside two controls, and inhibition was derived from the corrected relative luminescence values. Validity was confirmed by calculating KF-values for both exposure times. The correction factor at 10 minutes (KF10 = 0.74) satisfied the ISO validity criterion of 0.6–1.8, while the 20-minute factor (KF20 = 0.53) fell outside this range, reducing the reliability of the later dataset. Concentration-inhibition curves were successfully plotted through nonlinear regression using a log-normal dose-response model in R.

Inhibition was observed to occur at higher concentrations (> 60 mg/L) reaching an inhibition average of 65.56% and 43.34% at 240 mg/L for the 10 and 20 minute respectively. At lower concentrations negative luminescence values were recorded indicating low-dose stimulation (hormesis). Based on the valid 10-minute data, EC10, EC20, and EC50 were estimated at 116.8 mg/L, 140.9 mg/L, and 201.8 mg/L respectively. While based on the 20-minute data EC10, EC20, and EC50 were estimated at 183.5 mg/L, 204 mg/L and 249.9 mg/L respectively. The values calculated for both time points were not found to be statistically different, as the 95% confidence intervals overlap.

The obtained results are approximately an order of magnitude higher than literature values, suggesting that *Vibrio fischeri* exhibits a lower sensitivity to 4-nitrophenol compared to the reference invertebrate species. This discrepancy highlights the importance of species-specific toxicity assessments when evaluating ecotoxicological risk. Future studies should employ a broader concentration range to enhance the precision and reliability of the EC50.

INTRODUCTION

The goal of this experiment was to quantitatively determine the acute toxicity of 4-nitrophenol's to the marine bioluminescent bacterium *Vibrio fischeri* by performing a short-term luminescent inhibition test according to ISO 11348-3 (using freeze-dried bacteria reconstituted in 20‰ NaCl and $NaHCO_3$ medium) [1].

Vibrio fischeri is a gram-negative, rod-shaped, motile marine bacterium that naturally produces visible bioluminescence. In ecotoxicology, V. fischeri is widely used as a standard test organism in rapid luminescence inhibition assays (e.g. Microtox and ISO 11348-based tests), where the degree of light inhibition after short-term exposure to water samples, effluents or single chemicals serves as a sensitive indicator of acute toxicity. In fact, the results make it possible to determine at which concentrations the contaminant begins to cause harmful biological effects, which can then be used to assess when the contamination may represent a potential environmental risk [2][3].

The compound 4-nitrophenol is characterized by high solubility (16 g/L at 25 °C), indicating that it is very mobile and tends to remain in the aqueous phase. On the other hand, it has low volatility since both Henry's constant and vapor pressure are small (H <10^-5 atm-m3/mol and vapor pressure <0.01 mmHg). As a result, the compound is not expected to partition into the gas phase [4][5]. This makes 4-nitrophenol a safe compound to work with in the lab since inhalation is not expected to be a significant exposure pathway. Moreover, losses of the compound are not expected, as its very low volatility ensures that it remains within the vials.

METHOD

The BIOTOX test has been conducted for five different concentrations of 4-nitrophenol mixed in a saltwater and bacteria matrix. The concentrations have been deducted from an initial EC50 value of 30 mg/L (average of literature findings [6][7]) and accordingly reduced or increased by a certain factor (¼, ½, 1, 2, 4) to have a meaningful range of test values.

- Concentration of the stock solution: 2400 mg/L
- Concentration range used: [0, 15, 30, 60, 120, 240] mg/L.

Each concentration was tested twice and to have a basis for comparison, and the test also included two control samples without any addition of the contaminant, making a total of 12 replicates.

All the replicates had a total volume of 200 µL that included 100 µL of bacterial suspension and 100 µL from the concentration series.

Raw luminescence data for t=0, t=10, and t=20 min were recorded (See *Table 4* in Appendix). For each exposure time, a correction factor was calculated as $K_{F_x} = \frac{IC_x}{IC_0}$, where IC_x is the mean luminescence of the two control vials after x minutes exposure and IC_0 is the mean control luminescence at t=0. Validity of each dataset was confirmed by checking that the calculated correction factors (KF) fell within the acceptable range of 0.6–1.8 [1].

The relative response (REL) for each test concentration and replicate was then computed as $Rel_x = \frac{IT_x}{K_{F_x} \times IT_0}$, where IT_x and IT_0 are the luminescence intensities of the test sample after x minutes exposure and at t=0, respectively. Inhibition of luminescence was derived as $100 \times (1 - Rel_x)$ and expressed as a percentage relative to the corrected control.

Concentration-response datasets (10 min and 20 min exposure) were analyzed using R software following DTU guidance (*Figure 1 & 2*).

RESULTS

For both exposure times, luminescence inhibition generally increased with increasing concentration, as can be seen from the values in *Figure 1 & 2.* However, large variability was observed at low and medium concentrations. Below 60 mg/L, inhibition values were negative or close to zero, indicating no clear inhibitory effect. A clear increase in inhibition was only observed at higher concentrations, where the curve showed a typical sigmoidal trend. At the maximum concentration (240 mg/L), the average inhibition percentages reached were 66% at t= 10 min. and 43% for t=20 min. This means that for t=20 min. the 50% inhibition was not reached.

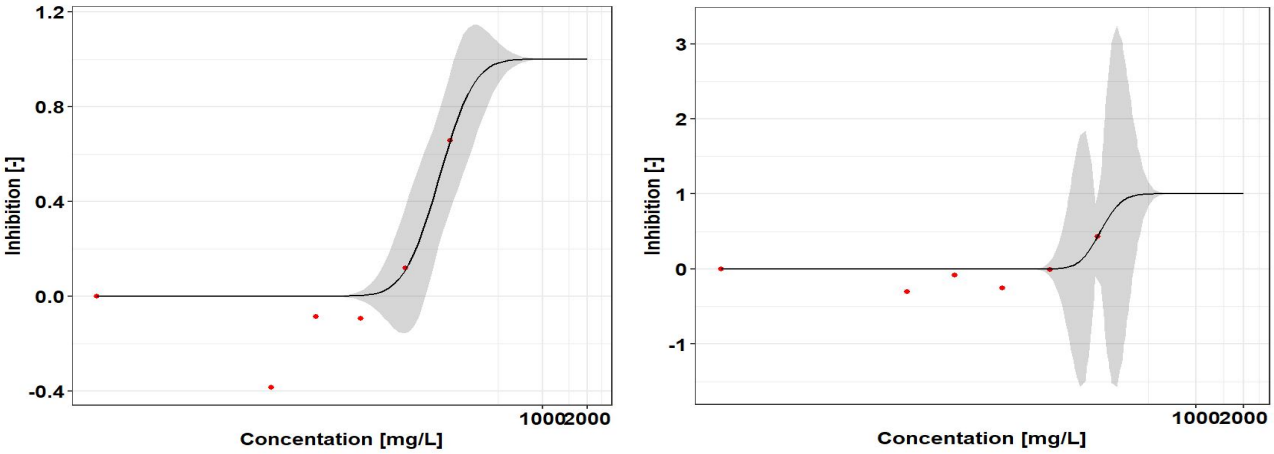


Figure 1, 2 concentration-inhibition curves for t=10 minutes (left) and t=20 minutes (right).

The effective concentrations (EC) estimated from the 10 min. exposure data were:

Table 2. EC10, 20 & 50 After 10 minutes.

	Value (mg/L)	SD	Lower (95% CI)	Upper (95% CI)
EC10	116.8	32.91	43.44	190.1
EC20	140.9	30.05	73.94	207.8
EC50	201.8	27.30	141.0	262.7

The EC estimated from the 20 min. exposure data were:

Table 3. EC10,20 & 50 After 20 min.

	Value (mg/L)	SD	Lower (95% CI)	Upper (95% CI)
EC10	183.5	186.9	232.9	599.9
EC20	204.0	126.2	-77.24	485.3
EC50	249.9	44.05	151.8	348.1

The EC10, EC20, and EC50 derived from the 10 min. exposure data are presented in *Table 2*, together with their standard deviations (SD) and 95% confidence intervals (CI). The results show a clear increase in EC values with increasing effect levels, indicating a consistent dose–response relationship. However, the relatively wide confidence intervals suggest a degree of uncertainty in the estimates, likely due to higher variability in the low-effect range of the dataset.

The corresponding EC values obtained from the 20 min. exposure are shown in *Table 3.* Compared to the 10 min. results, these EC values are generally higher and are associated with substantially wider confidence

intervals, in some cases even including negative lower bounds. This indicates reduced precision and reliability of the estimates at this exposure time. Notably, this trend contradicts theoretical expectations, as longer exposure times are typically associated with increased toxicity and therefore lower EC values. The observed increase in EC values at 20 min may be attributed to experimental variability, insufficient data in the effective concentration range, or instability in the model fitting.

According to ISO 11348-3 (2007) [5], which specifies quality criteria for luminescence inhibition tests using *Vibrio fischeri*, the validity range for the KF value is defined as 0.6–1.8 [1]. In this study, KF10 = 0.74 met this criterion, whereas KF20 = 0.53 fell below the acceptable range. Therefore, the 10 min dataset was considered reliable and used as the primary basis for estimating EC values, while the 20 min results were used only for comparison and interpreted with caution.

Other observations

1. At several low and medium concentrations, the calculated inhibition values were negative, and such negative inhibition values may occur due to experimental variability or slight stimulation of bacterial activity at low concentrations. Overall, the data show a clear response only at the high concentrations tested.
2. Although the estimates showed relatively large standard errors and wide confidence intervals, the confidence intervals for EC10, EC20, and EC50 were generally comparable in magnitude. This indicates that, despite the variability in the data, the model provided reasonably consistent estimates of the effective concentrations.
3. When making the stock solution, it was observed that the use of a magnetic bead and an agitator were needed to fully dissolve the amount of 4-nitrophenol. This was unexpected considering the compound's high solubility, as previously mentioned (16g/L at standard conditions).

DISCUSSION

Assay validity depends on the correction factor (KF), which should fall between 0.6 and 1.8. While the 10-minute KF (0.74) met this criterion, the 20-minute KF (0.53) fell slightly below it but is still analytically usable (cautious interpretation is needed).

Overall, the test was conducted without major complications, but pH adjustment of the stock solution was required since it was deviating more than 1 unit from the medium (pH=8). The measured value was 6.4 which had to be corrected with the addition of NaOH to reach a more alkaline pH (the value had to fall within 7 and 9) [1]. The exact volume added was not quantified and considered negligible, but this may have influenced the results. This value of pH paired with a pKa value of 7.15 leads to a halfway behavior regarding 4-nitrophenol's dissociation properties (half neutral and half ionized). Sorption can, for this reason, considered to be a minor factor in the results precision. In fact, 4-nitrophenol has moderate sorption behavior so some interaction with the vials walls is possible, but most of the compound remains dissolved and bioavailable [8].

As for the results, what could be observed is that at concentrations below approximately 60 mg/L, 4-nitrophenol didn't inhibit the bacteria reflected in the negative values in Figure 1 & 2. This is a known effect called hormesis, and it likely happened because our selected concentration range primarily captured the compound's 'low-dose stimulation' zone not showing clear toxicity until higher concentrations [9].

Furthermore, a full concentration-response curve (spanning from 0% to near 100% inhibition) was not achieved due to the low concentration range. While the valid 10-minute data set can be used to estimate EC values, the estimations carry considerable uncertainty. In fact, the results showed large standard errors and very wide confidence intervals. This is due to the statistical model lacking sufficient data points at higher concentrations to anchor the upper plateau of the inhibition curve.

To address the lack of high-inhibition data points and stabilize the model, an extrapolation approach was applied by artificially introducing a theoretical concentration of 1000 mg/L into the dataset. The reasoning behind this is biologically sound: at such extreme concentration, the metabolic processes of Vibrio fischeri would be completely halted, leading to near 100% luminescence inhibition and creating a biological "plateau." By feeding this plateau data point into the model, we effectively anchored the upper asymptote of the dose-response curve. This mathematical extrapolation forced the model to complete the sigmoidal shape, which significantly reduced the standard errors and narrowed the initially wide confidence intervals of our EC50 estimation.

Even though the EC values are not equivalent between the two intervals, they cannot be considered statistically different since the confidence intervals are overlapping (see Table 2 & 3). The accuracy of the EC values is not affected by the physical-chemical properties. In fact, 4-nitrophenol is a compound that is characterized by low volatility and high solubility, making it stay dissolved within the aqueous phase in the vials [4]. Therefore, the prepared concentration remains unchanged during the test and the bacteria get homogeneous and stable exposed. This also increases confidence in the EC values as volatility correction factors don't have to be considered and avoid underestimations.

Comparison with Literature EC-values for 4-Nitrophenol

According to the EPA Hazard Characterization and ANZ Guidelines data collected in our previous sessions, the acute toxicity of 4-nitrophenol across different trophic levels is relatively high. The 48-h EC50 for aquatic invertebrates is 22 mg/L [10], and the 72-h/96-h EC50 for aquatic plants (e.g., Desmodesmus subspicatus) is > 32 mg/L [11]. When comparing our calculated 10-min Vibrio fischeri EC50 of 201.9 mg/L to these literature reference values, a massive discrepancy is evident: our experimental result is nearly an order of magnitude higher, severely underestimating the true toxicity of 4-nitrophenol. This discrepancy is probably because we are comparing EC50 values from bacteria to EC50 values referring to higher trophic level organisms.

Furthermore, the extrapolated EC-values from bacteria cannot be safely compared to standard trophic-level data (algae or invertebrates) for environmental risk assessment since it failed to capture the true toxic threshold and the full inhibitory phase is not reached.

CONCLUSION

The luminescence inhibition test with *Vibrio fischeri* demonstrated that the acute toxicity of 4-nitrophenol increased with increasing concentration within the tested range of 7.5–240 mg/L. Little or no inhibition was observed at low and medium concentrations (≤ 60 mg/L), where several negative inhibition values occurred, likely due to experimental variability or low-dose stimulation (hormesis). A clearer inhibitory effect appeared only at the highest concentrations tested.

The most reliable results were obtained from the 10-minute exposure data, as the correction factor met the ISO validity criteria, while the 20-minute dataset showed reduced reliability. Based on the valid 10-minute dataset, the estimated EC10, EC20, and EC50 values were 116.8 mg/L, 140.9 mg/L, and 201.8 mg/L, respectively. These estimates were associated with relatively wide confidence intervals, reflecting variability in the dataset and the limited number of concentrations within the effective response range. The 20-minute dataset was considered less reliable due to an invalid correction factor but due to the small deviation from the KF range, results were still analyzed providing EC10, EC20, and EC50 values of 183.5 mg/L, 204.0 mg/L and, 249.9 mg/L respectively.

Therefore, the results indicate moderate acute toxicity of 4-nitrophenol toward *Vibrio fischeri*, but they likely underestimate the compound's toxicity because the experiment did not fully capture the upper plateau of the inhibition curve. Overall, the study highlights the importance of selecting an appropriate concentration range and ensuring sufficient data points across the full dose–response curve to obtain more precise toxicity estimates.

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APPENDIX

Raw data table Excel

Table 4. Raw data and REL & inhibition calculations.

Test bottles	Conc. (mg/L)	t=0	t=10min	t=20 min	REL (10 min)	REL (20 min)	% Inhibition t=10min	% Inhibition t=20min
1.a (control)	0	66878	49505	36562	1	1	0.00	0.00
1.b (control)	0	61325	45966	31626	1	1	0.00	0.00
2.a	15	54613	56386	38641	1.386442841	1.330278459	-38.64	-33.03
2.b	15	58177	59929	39677	1.383287413	1.28226486	-38.33	-28.23
3.a	30	63675	50788	36006	1.071072512	1.063153955	-7.11	-6.32
3.b	30	61337	50268	35990	1.100514558	1.103188056	-10.05	-10.32
4.a	60	63141	53184	44458	1.131087645	1.323819278	-13.11	-32.38
4.b	60	66882	52719	41951	1.058484807	1.179297293	-5.85	-17.93
5.a	120	65434	44570	37864	0.914673045	1.087960824	8.53	-8.80
5.b	120	60150	37949	29843	0.847210678	0.932818354	15.28	6.72
6.a	240	27278	5784	7195	0.284736063	0.495916443	71.53	50.41
6.b	240	58866	17708	19951	0.403953805	0.637221438	59.60	36.28

Concentration- response table to plot in R & calculated average response

Table 5. Concentration-response table to plot in R.

Test bottles	T= 10 min.			T= 20 min.		
	Conc	Resp	Resp. Average	Conc	Resp	Resp. Average
1.a (control)	0	0.00		0	0.00	
1.b (control)	0	0.00	0.00	0	0.00	0.00
2.a	15	-0.39		15	-0.33	
2.b	15	-0.38	-0.385	15	-0.28	-0.305
3.a	30	-0.07		30	-0.06	
3.b	30	-0.1	-0.085	30	-0.10	-0.08
4.a	60	-0.13		60	-0.32	
4.b	60	-0.06	-0.095	60	-0.18	-0.25
5.a	120	0.09		120	-0.09	
5.b	120	0.15	0.12	120	0.07	-0.01
6.a	240	0.72		240	0.5	
6.b	240	0.6	0.66	240	0.36	0.43
n.a.	1000	n.a.	n.a.	1000	n.a.	n.a.
n.a.	2000	n.a.	n.a.	2000	n.a.	n.a.

Results (code of concentration-response curves / EC values)

```
#Script for estimation of concentration-response relationships for course 12237

rm(list = ls())#delet everything from Environment
library(tidyverse)##To use the pipe (%>%) and ggplot
library(drc)
library(readxl)##To import excel to R
library(skimr)##To summarise your data


#++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+
#Mark your data in excel in two columns named Conc and Resp
my_data2 <- read.table(file = "clipboard",
                       sep = "\t", header=TRUE)


conc <- dplyr::pull(my_data2, Conc)

resp <- dplyr::pull(my_data2, Resp)
```

```
#++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+
#Check and view your data
my_data=tibble(con=conc,
                        resp=resp)
#++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+++
# 1)Plot in ggplot
my_data%>%#your data
   ggplot(aes(x=con,y=resp))+
   geom_point(size=4)+
   labs(x="Concentration", y="Response")+#names of the axis
   theme_bw()#change the appearance of the graph
#++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+++
#++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
+++
#2) Set the model using drm function
fit <- drm(
   formula = resp~con,
   data = my_data,
   fct = LN.2()#
)
#++++Plot using base
R++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
#3) Plot the model using base R
plot(fit)
#++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
++++++++++++++
#4)
#++++4.1 New function
##For prediction purposes, we have to build a new dataset
#We use sequence function, which i explained as follow: seq(from, to, number of observation)
newdata <- expand.grid(conc=exp(seq(log(1), log(max(my_data$con)), length=100)))
#For this new data, we used: seq(log(min(my_data$con), log(max(my_data$con)), length=100))
#To avoid log(0), which does not exist, we set the min to 0.5.

#++++4.2 Prediction dataset
#Now we make the prediction
pm <- predict(fit, newdata=newdata, interval="confidence")
```

```
#++++4.3 Combine new dataset and the prediction dataset
my_pm <- newdata%>%
   bind_cols(#Combine many columns
      pm%>%as.data.frame()##transform pm to data frame##
   )
#check your new dataset
my_pm%>%head()

#++++4.4 Your lovely graph is here:
my_data%>%#our original data
   mutate(Concentration=ifelse(con==0,1,con))%>%
   #we create a new variable called Concentration
   #we say that if con=0, then the new variable should have 0.5,
   #else keep the value of conc
   #This is equivalent to: my_data$conc[my_data$con == 0] <- 0.5
   ggplot(aes(x = Concentration, y = resp)) +
      stat_summary(geom = "point", fun.y = "mean", col= "red") +
   geom_ribbon(data=my_pm, aes(x=conc, y=Prediction, ymin=Lower, ymax=Upper), alpha=0.2) +
   geom_line(data=my_pm, aes(x=conc, y=Prediction)) +
   coord_trans(x="log") +
   xlab("Concentation [mg/L]") + ylab("Inhibition [-]")+theme_bw()+
   theme(axis.title=element_text(size=14,face="bold"))+
   theme(axis.text.x = element_text(face="bold", color="black",size=14),#, angle=90),
            axis.text.y = element_text(face="bold", color="black",size=14),
            strip.text= element_text(face="bold", color="black",size=14)) +

#+++
#Table for EC-values 10, 20, 50
ED(fit, c(10,20,50), interval = "delta")%>%
   View()#To show it in view
```